Brian Wonjun Lee

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Education

University of Pennsylvania Philadelphia, PA
MSE in Computer Graphics & Game Technology (CGGT), Submatriculation *Expected May 2028*
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars *Expected May 2027*
Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) *June 2026 – Present · San Francisco (Remote)*

- Shipped a production retention system (targeted push via Supabase, Deno edge functions, FCM) with a treatment/holdout A/B, **recovering churned users back into paying subscribers**
- Expanded reachable users **~2.7×** by validating the win-back targeting against live production data and correcting how eligible users were identified
- Turned the company's production AI agent into a drop-in package other teams can embed in their own apps — a typed SDK with versioned APIs and **136 automated tests**; a new AI coding agent integrated it into a fresh app on the **first attempt**
- Diagnosed and fixed a production authentication outage (a JWT signing-key migration) that had silently broken the notification pipeline, restoring it **from failing (401) to healthy (200)**

Software Engineer Intern, BitMango (Puzzle1 Studio) *June 2025 – Aug 2025 · Pangyo, South Korea*

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet evolving app-store requirements and sustain production availability

Project Lead & Product Designer, Penn Spark *Jan 2025 – Present · Philadelphia, PA*

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army) *Dec 2022 – June 2024 · Seoul, South Korea*

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm *June 2022 – Aug 2022 · Seongnam, South Korea*

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

Projects

Mini Minecraft | *C++, OpenGL, GLSL* *Apr 2026*

- Built **7-biome procedural world generation** for a from-scratch voxel engine (team of 3) — Perlin/FBM/Voronoi noise with smoothstep blending
- Carved **3D Perlin-noise caves** and post-process water/lava overlays via a framebuffer pass
- Engineered the GLSL **render pipeline** — PCF shadow mapping, screen-space reflections, vertex AO, and a day-night cycle

Passenger | *Unreal Engine 5, HLSL, Maya* *Mar 2026 – Apr 2026*

- Wrote 2 **HLSL post-process shaders** (anisotropic Kuwahara, cell shader) sampling GBuffer channels for stylized rendering
- **Rigged and animated a character** in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint
- Rendered at 2560×1080 via Movie Render Pipeline with **hardware ray tracing** over Lumen GI, 64 TSR samples/frame

Mini Maya | *C++, OpenGL, Qt* *Feb 2026 – Mar 2026*

- Implemented a **half-edge mesh data structure** (Face/HalfEdge/Vertex, smart-pointer ownership) for topology-aware traversal and editing
- Wrote an **OBJ parser** that builds the mesh from arbitrary n-gon inputs, rendered via OpenGL VBOs
- Implemented **Catmull–Clark subdivision** and topology ops (split, extrusion, triangulation) in a Qt editor

Technical Skills

Graphics & Rendering: OpenGL, GLSL, HLSL, shader programming, Unreal Engine 5, Lumen, real-time & offline rendering
3D & Tooling: Maya, Blender, Substance Painter, Qt, Movie Render Pipeline
Languages: C++, Python, TypeScript, JavaScript, Java, C#, GLSL/HLSL, SQL
Frameworks & Tools: React Three Fiber, Three.js, React, Node.js, Git, GitLab, VS Code